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Enhancing IoT Security Using Machine Learning for Port Scan Attack Detection

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Abstract

The fast proliferation of the Internet of Things (IoT) has facilitated the highest levels of connectivity, but also has created severe weaknesses, especially reconnaissance attacks, including host and operating system (OS) port scanning. Such attacks provide gateway to even more sophisticated attacks, and therefore it is crucial to detect them early to ensure IoT infrastructures are secured. This paper suggests a framework that could be used to detect port scanning based on the IoTID20 dataset and implemented with machine learning. Eight classifiers, including, XGBoost, Random Forest, Bagging, AdaBoost, Gradient Boosting, CatBoost, LightGBM and Support Vector Machine (SVM) have been used in comparative analysis in terms of Accuracy, Precision, Recall, and F1-score. It is demonstrated that the boosting and ensemble models are much better than the traditional classifiers, with LightGBM demonstrating the best accuracy of 99.90%, precision of 99.91%, recall of 99.87% and F1-score of 99.89%. Near-perfect detection was also observed with bagging (99.87% accuracy) and CatBoost (99.84% accuracy) and the lowest accuracy was observed with SVM (98.37%). These results highlight the power of sophisticated ensemble learning models to identify complex attack patterns in high-dimensional IoT traffic. The combination of these strategies can offer a very strong backbone to the next generation intrusion detection systems that will be able to overcome reconnaissance threat in the IoT networks

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1. Introduction

Internet of Things (IoT) has become one of the most groundbreaking technologies of the past decade that brings together billions of devices in the context of smart houses, medicine, industry, and transportation [1-5]. This unparalleled connectivity creates enormous volumes of information and allows making smart decisions, but it also dramatically increases the attack surface of cyber-attackers [6]. IoT devices are typical resource constrained systems unlike traditional computing systems, which may operate lightweight operating systems and are deployed in heterogeneous environments with little security hardening [7, 8]. Consequently, the IoT ecosystems are very appealing targets to cybercriminals who want to compromise the devices, steal confidential information, or develop large-scale attacks like botnets and distributed denial of service (DDoS) [9-11].

The reconnaissance activities which include port scanning for host and OS are among the first stages in the cyberattack lifecycle [12, 13]. Through scanning a network to determine open ports and OS features, attackers can collect important intelligence that can help them discover vulnerable services and customize their future exploits. Port scans and OS scans are cheap to launch and can be done in stealthy or distributed modes that mimic benign traffic and are thus especially hard to do in IoT networks [14]. If it goes unnoticed, reconnaissance opens the door to malware injection [15, 16], privilege escalation, and long-term intrusions potentially crippling the IoT infrastructure. Therefore, prompt and correct identification of host and OS port scan attempts is a necessary defense mechanism when securing the IoT systems.

Conventional intrusion detection systems (IDS) and signature-based surveillance tools are not effective in reconnaissance in the IoT setting. Signature based methods are based on familiar attack patterns and cannot work with zero-day or sophisticated scanning methods. The threshold-based method, like solutions with flag connections above a specific number of probes, tend to have high false positives because of the normal IoT activity of firmware updates or device discovery. As a result, the necessity to develop intelligent, adaptive and data-driven mechanisms of detection, which will help to isolate reconnaissance and normal IoT traffic with greater accuracy and reliability grows.

Machine learning (ML) is part of artificial intelligence [17, 18], it has demonstrated a lot of potential in this field and is able to learn complicated patterns based on network traffic and generalize to an unseen attack behavior [19, 20]. Ensemble classifiers provide a form of robustness through aggregating multiple learners to attain better forecasting strength and less variance [21, 22]. Some studies have already investigated the use of ML to perform intrusion detection on IoT [23, 24], although much of the earlier research has concentrated on volumetric attacks like DDoS, reconnaissance detection has not yet been studied extensively. Moreover, very little comparative analysis has been conducted on a large collection of ensembles and boosting algorithms on recent IoT data, and it remains unclear on which classifiers offer the best results in port and OS scan detection. To fill these gaps, this paper proposes full machine-learning architecture to identify host and OS port scan attacks in IoT settings. In this paper, we test a wide range of classifiers with the use of the IoTID20 dataset, which includes labeled IoT traffic data with reconnaissance cases. They are Random Forest [25], Bagging, AdaBoost, Gradient Boosting, CatBoost, LightGBM, XGBoost, and Support Vector Machine (SVM) [26]. Their ability to learn non-linear decision boundaries and work with imbalanced datasets is anticipated to be taken advantage of by ensemble and boosting methods which can be compared to a classical baseline of SVM [27]. We evaluate the models based on common indices like accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, and especially in terms of their capacity to reduce false alarms and reliably identify reconnaissance. This paper has made major contribution which is comparative analysis among eight machine learning classifiers with special focus on ensemble and boosting algorithms to find their success in Scan Port detection.

The remaining part of the paper is structured in the following way: Section II is the related work review, Port Scan attack detection model, and ML-based IoT security. Section III explains the suggested methodology, such as data collection, and model design. Section IV shows the experimental findings and performance analysis. In section V, there are findings and implications. Lastly, the paper ends with Section VI, which summarizes the future research direction.

2. Related works

This fast development of IoT ecosystems has attracted immense research on intrusion detection, especially the reconnaissance stage where attackers use port scanning and OS fingerprinting to acquire information about devices

that are vulnerable. There are many studies that have used machine learning and more recently ensemble and deep learning to identify such activities [28]. In this section, the major contributions in the field are reviewed, their datasets, methods, and findings are noted, and their limitations are described, which inspires our efforts.

In their article, Algaolahi et al. [29] suggested a guided machine learning-based method in identifying port-scanning attacks through the CICIDS2017 dataset, where the authors concentrated on such classifiers as Decision Tree, Random Forest, AdaBoost, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Their findings indicated that Decision tree, Random Forest, and KNN had almost perfect detection rates with an accuracy and F1-score of 99.8 and above and that AdaBoost was slightly lower with 89.61 accuracy. They have also added the concept of computational efficiency and thus established that Decision Tree provided the best balance between processing time and detection accuracy. This work has shown how tree-based supervised approaches could be effective in the detection of reconnaissance operations but was constrained by conventional algorithms and one dataset. Conversely, our work builds upon this research by testing a larger collection of modern ensembles and increasing the number of classifiers, such as LightGBM, CatBoost, and XGBoost, on the IoTID20 dataset that has been decidedly tailored to the environment of the IoT.

Abu Al-Haija et al. [30] proposed a predictive model to identify port scanning attacks based on the PSA-2017 data, which includes over 700,000 traffic packets with the following characteristics: throughput, the frequency of port visit, and source/destination IP addresses. The authors compared six supervised learning algorithms, which were logistic regression, decision trees, discriminant classifiers, naive Bayes and ensemble boosted trees and used five-fold cross-validation to guarantee robustness. Their findings showed that logistic regression performed significantly better than the rest with an accuracy of 99.4, precision of 99.9, recall of 99.4 and F1-score of 99.7, and a very low detection overhead of 0.454 us. This work has shown the unexpected performance of what would otherwise be considered a relatively simple model like logistic regression when a carefully designed set of features is used to train it. Nevertheless, it was only tested on one dataset and not more sophisticated models that use boosting like LightGBM or CatBoost. Our project is based on this premise by applying a wider range of ensemble and gradient boosting models, trained on the IoTID20 dataset, to give closer insights into the applicability of modern classifiers to reconnaissance detection in IoT networks.

Aamir et al. [31] did a mass comparative study of 22 machine learning algorithms in identifying port scanning and DDoS attacks using the CICIDS2017 dataset which contains more recent attack vectors and realistic traffic conditions. Extensive preprocessing was used in the study including feature selection using correlation and normalization, and training models, including decision trees, discriminant analysis, KNN, SVM, and ensemble classifiers. Their findings demonstrated that the Fine Gaussian SVM became the best with a high accuracy of 99% whereas all the variations of discriminant analysis and SVM all had a consistently high accuracy of 90% in testing. Interestingly, most tree based, and ensemble models did not perform well with the highest accuracies ranging between 49 and 69 indicating their weaknesses in this dataset. The authors also confirmed their results with 10-fold cross-validation, ROC-AUC, and dimension reduction with PCA and found that SVM and discriminant analysis performed the best in port scan and DDoS detection. Nevertheless, the lack of contemporary boosting methods, including LightGBM, XGBoost, and CatBoost, restricted the research. We build on this work by including these sophisticated models and testing them on the IoTID20 dataset, which is specifically created to characterize the IoT traffic.

Kumari and Sharma [32] suggested using the Aposemat IoT-23 dataset, including more than 600,000 samples of benign and malicious traffic, to propose an AI-enabled detection framework to secure the IoT networks against the port scanning attack. The combination of their approach was the use of Convolutional Neural Networks (CNN) and the use of classic machine learning models (Naive Bayes and Logistic Regression) to detect reconnaissance traffic in real time. The findings revealed that CNN was able to perform better than the other classifiers, with higher accuracy, precision, recall, and F1-score since it was able to learn the more intricate patterns in IoT traffic. The paper emphasized the potential of deep learning, in particular, CNN, in the context of IoT security, although it has been largely limited to one model of deep learning and two comparison methods of ML approaches, without considering more sophisticated ensemble boosting approaches. Our work, in its turn, is a complement to this body of research by comparing a wider range of state-of-the-art boosting and ensemble classifiers (LightGBM, CatBoost, XGBoost, etc.) against the IoTID20 dataset and thus contributes to a more comprehensive comparative analysis of detecting port scanning attacks in IoT stakes.

Altidor and Talhi [33] suggested an integrated framework based on the combination of Genetic Algorithms (GA) with machine learning classifiers to increase the rate of port scanning along with DDoS attacks detection. Selecting features and optimizing hyperparameters were used, where the GA was used to reduce the dimensionality and increase the accuracy of classification. Their tests were performed on three benchmark datasets, CICIDS2017, CICIOT2023 and MSCAD, which involved traditional enterprise networks, IoT traffic, and multi-stage cyberattacks. Findings indicated that GA-based optimization yielded much higher performance in detection than Random Forest and XGBoost reached almost perfect detection (as high as 99.99% accuracy) on CICIDS2017 and consistently high detection (over 95% accuracy) on IoT-oriented CICIOT2023 and MSCAD datasets. Although very useful, the research noted the great computation requirements of GA and the possibility of over-fitting due to over-dependence on the most significant factors like destination port numbers. This study has demonstrated the promise of metaheuristic algorithms in the IDS but primarily in optimization pipelines. Conversely, our work focuses on a comparative assessment of various boosting and ensemble classifiers on the IoTID20 dataset directly, and the results insight into real-time IoT port scan detection without the need to employ costly metaheuristic optimization.

Yan et al. [34] suggested a port scanning detection tool using ensemble learning to eliminate the drawbacks of individual machine learning. They used a combination of Decision Tree, Support Vector Machine (SVM), Naive Bayes and Logistic Regression as the base classifiers and a majority voting strategy to combine their prediction. The model was tested on the CICIDS2017 dataset, and a self-collected dataset and its performance was evaluated based on accuracy, precision, recall, F1-score, false alarm rate, and missing alarm rate. The ensemble method was found to have high accuracy of 98.83, F1-score of 99.31 and precision of 99.93 and was superior to most of the individual classifiers, as well as minimizing false alarms as opposed to standalone models. Moreover, feature importance showed that there are ten very sensitive features (i.e., ACK flag count, flow duration, packet length mean) that greatly define port scanning behavior. In this study, the potential of ensemble methods was proved, but with rather simple base models, and our research is the next step in this direction, as it evaluated modern boosting ensembles, including LightGBM, CatBoost, and XGBoost, on the IoTID20 dataset, which can be used to better understand the principles of reconnaissance detection in the IoT.

Akif et al. [35] created an IoT intrusion detection system, which was based on an adapted version of the Random Forest algorithm, and is aimed at addressing some of the limitations of the conventional ensemble classifiers. The suggested changes involved using one-hot encoding of categorical targets, a more economical approach to bootstrapping with less redundancy, a combined feature selection process and an evaluation phase in decision trees to enhance the precision of classification. Three popular benchmark datasets were used to validate the model: Iootid20, UNSW-NB15 and IoT-23. The experimental results indicated that the modified Random Forest achieved 96.2, 98.85 and 99.93 percent accuracy in the test of Iootid20, UNSW-NB15, and IoT-23 respectively, and were better than the performance of the traditional classifiers used in the experiment, including logistic regression, decision trees, and gradient boosting. Moreover, the recall, precision, and F1-scores remained over 95% which validates the strength of the method. Although the method was better at detecting attacks in all datasets, the authors recognized that scaling this approach and adapting to new emerging attacks, which necessitate retraining, is problematic. The research shows the usefulness of better ensemble techniques in detecting IoT intrusion, although it primarily concentrates on the improvements of the use of the Random Forest, and it remains open to research other more advanced boosting strategies. In this spirit, our work will create models of advanced boosting (LightGBM, CatBoost, XGBoost) on the IoTID20 dataset to perform reconnaissance detection in a more comprehensive and scalable way.

To conclude, the available literature on the problem of port scan detection in IoT and other similar networks has proven the efficiency of supervised machine learning, ensemble algorithms, and even deep learning in detecting reconnaissance attacks. Past research has concentrated on classical classifiers like decision tree, SVM, and logistic regression, where they usually achieve good performance, but only in a single dataset and small feature bases. Later literature has investigated ensemble methods, and hybrid methods, such as boosting algorithms, metaheuristic optimization, and deep learning, which are highly accurate, but can be very complex, unable to generalize, or computationally expensive. With these developments, however, there is also evident research gap to systematically assess state-of-the-art boosting ensembles (LightGBM, CatBoost, XGBoost, etc.) on more representative of heterogeneous IoT traffic datasets, such as the Iootid20 dataset. In response to this gap, our research builds on the previous literature by offering a detailed comparative study of modern ensemble learning algorithms to conduct host and OS port scan detection, revealing their applicability to use in real-world IoT security systems.

3. Port Scan Detection Model

In this section, the author explains the methodology used to improve the security of the IoT by identifying the attacks of port scanning of hosts and operating systems with the help of machine learning. Figure 1 shows the proposed framework, and it has a logical flow beginning with the selection of the dataset and ending with the assessment of the model. All these stages will be discussed below.

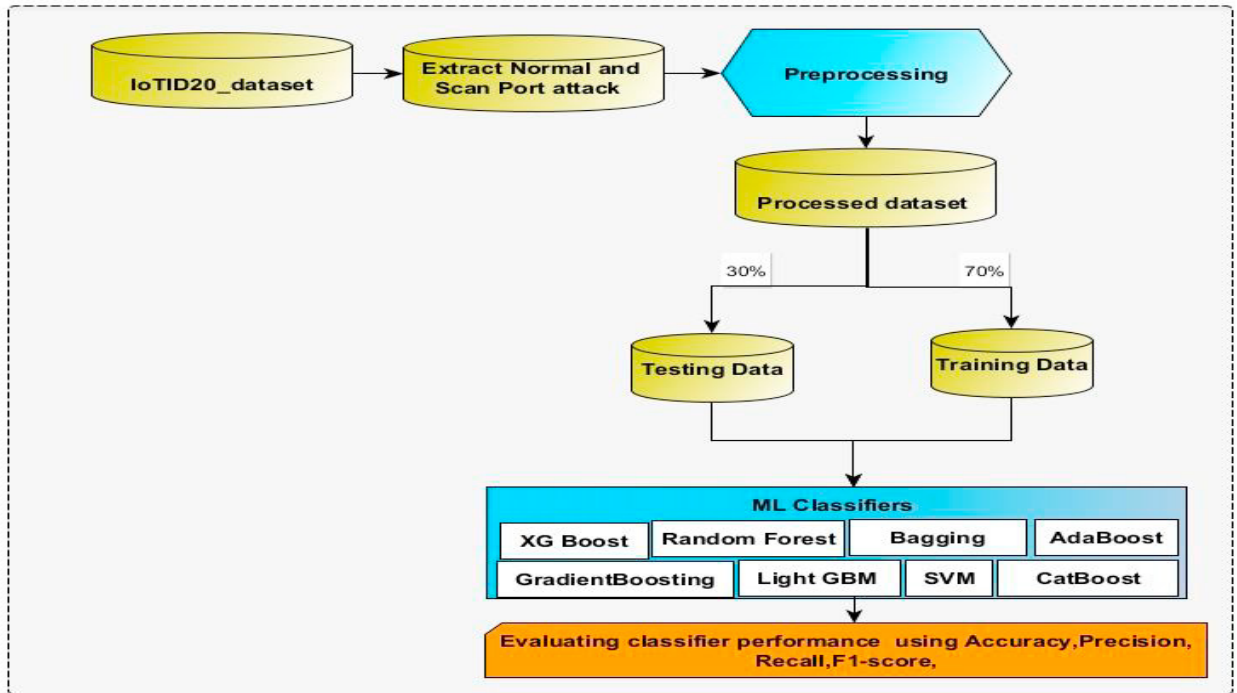


Fig.1: Port Scan Detection Model

3.1. Dataset Preprocessing

This work uses the IoTID20 dataset [36], which is a publicly accessible dataset that is explicitly engineered to be used in intrusion detection in Internet of Things (IoT) contexts. IoTID20 consists of network traffic generated by the IoT devices in diverse attack conditions such as Mirai, DOS, Scan and , MITM ARP Spoofing. This dataset is chosen as it provides more realistic IoT traffic patterns and a broad spectrum of malicious behaviors, thus being very appropriate to test machine learning-based intrusion detection systems [37].

This study targets normal traffic and port scan attack traffic. The reason behind this choice is that reconnaissance is usually the initial step in a cyberattack wherein attackers can scan IoT devices and determine their vulnerabilities. The framework can be used as a proactive defense mechanism by identifying port scans at an early stage. It is thus defined as a binary classification problem, and the task is to determine network traffic as either a benign or a port scan reconnaissance. Several preprocessing steps are implemented to guarantee the quality and compatibility of the data with machine learning models. These steps are Data Cleaning , Feature Normalization [38, 39], Categorical Encoding , and Dealing with Class Imbalance: This step produces a processed data set which is balanced, clean and is prepared to be classified.

3.2. Machine Learning Classifiers

Eight ML classifiers are used to measure the high effectiveness of identifying port scan attacks: XGBoost, Random Forest [40], Bagging, AdaBoost , Gradient Boosting, CatBoost, LightGBM, and SVM [41]. These classifiers were

selected because the performance of ensemble learning and boosting techniques is proven to be better than the performance of the SVM in the sphere of intrusion detection, and SVM represents a good basis in binary classification [42] [43].

3.3. Performance Evaluation

To evaluate the ability of the classifiers, the following measures are used [44]:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

$$\text{F1-Score} = \frac{2 * \text{Precision} * \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (4)$$

These metrics will be used to ensure that the models are properly assessed in all aspects, both regarding reliability and ability to distinguish normal and malicious IoT traffic.

4. Results and discussion

Accuracy, Precision, Recall and F1-score were used to evaluate the performance of the eight ML classifiers. These measures give a complete evaluation of the detection capability including overall correctness (accuracy), the ability to predict true attacks (precision), the ability to detect real attacks (recall), and a trade-off between the ability to predict and detect (F1-score). Table 1 shows the outcomes of the experiment

Table 1. Results

Classifier	Accuracy	Precision	Recall	F1-score
XGBoost	0.998	0.998	0.998	0.998
RandomForest	0.997	0.998	0.996	0.997
Bagging	0.999	0.999	0.999	0.999
AdaBoost	0.995	0.996	0.994	0.995
GradientBoosting	0.997	0.998	0.996	0.997
CatBoost	0.998	0.999	0.998	0.998
LightGBM	0.999	0.999	0.999	0.999
SVM	0.984	0.986	0.978	0.982

Accuracy refers to the percentage of instances of traffic that are well classified, which is a measure of the overall reliability of the model. LightGBM was the most accurate with 99.90, followed by Bagging (99.88%), CatBoost (99.84%), and XGBoost (99.83) in that order. These ensemble and boosting-based classifiers were shown to have high abilities in breaking the non-linear relationships that were complex in the IoT traffic. The performance of the tree-based methods (Gradient Boosting (99.74%) and Random Forest (99.73)) was also strong. AdaBoost achieved 99.53%, which is good but a little lower than other boosting methods. The lowest accuracy of the SVM model of 98.38 is still high, and it suggests that this method has certain weaknesses regarding the ability to process the volume and complexity of the traffic of the IoTID20.

Precision measures the number of the predicted port scan attacks that were correct and this is significant to reduce false alarms. LightGBM once again topped it by a large margin of 99.92% precision showing that they can be used to make very reliable predictions. This was superseded by Bagging (99.88%), CatBoost (99.87%), and XGBoost (99.84%), which are characterized by very high degrees of trust, when it comes to attack prediction. Gradient Boosting (99.79%), and Random Forest (99.76%) were a little lower and still great. AdaBoost achieved 99.57 which is not bad but still below the others. SVM had the least accuracy of 98.60 which is still good but relatively weak in comparison to ensemble learners.

Recall determines the percentage of real port scan attacks that have been detected and is important when intrusion detection systems cannot afford to miss attacks. LightGBM had the best recall of 99.87%, indicating almost perfect detection. Close behind was bagging (99.85%), XGBoost (99.80%), and CatBoost (99.79%) which also proved to be effective in the task of capturing reconnaissance behavior. Gradient Boosting (99.65) and Random Forest (99.65) were also good with slight decreases only. AdaBoost had 99.40 percent, which is a little less sensitive. The lowest recall at 97.83 was with SVM, which therefore is missing comparatively more port scan attempts than the ensemble models.

The F1-score is a tradeoff between precision and recall, which gives a general indication of the performance of the classification. LightGBM was again on top with 99.90 percent, which proves its better balance between good predictions and overall detection. Bagging (99.87%), CatBoost (99.83%), and XGBoost (99.82 %) were almost perfectly balanced. Gradient Boosting (99.72%) and Random Forest (99.70) are very consistent and AdaBoost (99.48) was weaker slightly. SVM had the poorest F1-score of 98.20, also competitive, but obviously less successful than current boosting algorithms in the tradeoff between detection and accuracy.

The relative outcomes show that the ensemble and boosting-based classifiers are evidently better than the traditional models in detection of the port scan of the IoT. LightGBM was the best model on all measures, making it the most stable model, closely surpassing Bagging, CatBoost, and XGBoost, which display great balance between detection and generalization. Gradient Boosting and Random Forest did not do very well but still were not as good as their optimized versions, whereas AdaBoost was behind because of its sensitivity to noisy and misclassified data. The lowest scores were found at SVM, and it emphasizes the drawbacks of margin-based classifiers in the context of high-dimensional and large-scale IoT traffic. In general, tree-based ensemble methods, especially LightGBM, CatBoost, and XGBoost, are the most advantageous to identify reconnaissance attacks in the IoT setting since they are resistant to the class imbalance and effective in modeling non-linear feature interactions. These results confirm that ensemble-based learning models offer a valid and feasible solution to host and OS port scan detection, and that they have a high probability of application in real-world IoT security structures requiring early reconnaissance detection to prevent large-scale attacks Figure 2 shows all classifiers performance comparison.

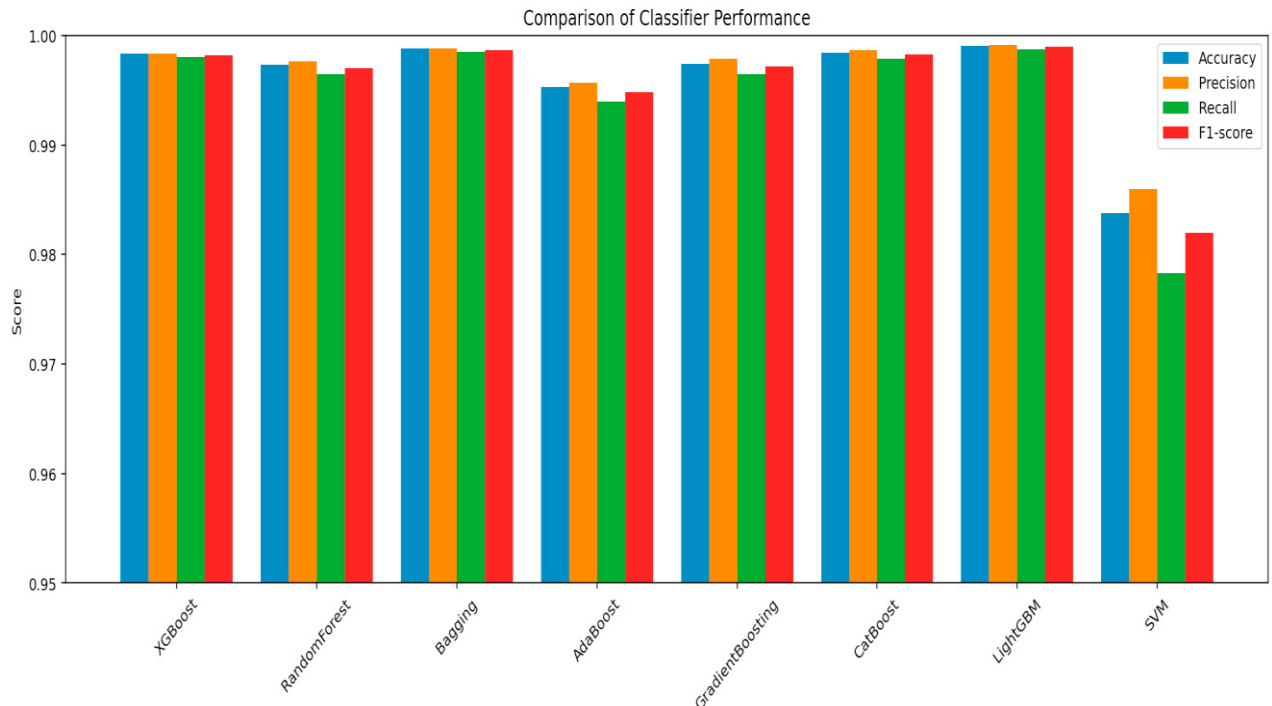


Figure 2 Classifiers performance comparison

5. Conclusion and future work

The research question that we were tackling in this study was to determine the efficacy of machine learning classifiers to identify host and OS port scans attacks in IoT settings using IootID20 dataset. Eight classifiers including XGBoost, Random Forest, Bagging, AdaBoost, Gradient Boosting, CatBoost, LightGBM and SVM were tested in four performance indicators, Accuracy, Precision, Recall, and F1-score. The findings were always consistent and showed that ensemble-based and boosting-based approaches were overwhelmingly better than the traditional models. LightGBM was the best among them with a close follow-up by Bagging, CatBoost, and XGBoost, which exhibited

better accuracy, precision, and strength in identifying reconnaissance operations. On the other hand, SVM had the worst performance, which represents a weakness in margin-based classifiers in high-dimensional internet traffic. These results prove that tree-based ensemble learning models are extremely effective in detecting port scans in IoT-based networks and can be a solid starting point in developing effective intrusion detection systems that can prevent early-stage cyberattacks. Although the findings are encouraging, this research also presents several research opportunities in the future. To further investigate how the evaluation captures time-dependent characteristics of the IoT traffic, first, the evaluation should be expanded to deep learning methods convolutional and recurrent neural networks. Second, the detection schemes might be hybridized to combine ensemble learning and deep feature extraction, which might be more resistant to adaptive port scan variations. Third, online learning and adaptive models should also be considered to provide real time detection in dynamic IoT systems. Lastly, future research can examine lighter implementations of these models to guarantee scalability and using them on resource-constrained IoT device implementations. By solving these directions, it will become possible to construct a stronger, more flexible, and scalable security framework that would enhance the resilience of IoT networks to reconnaissance, as well as other, newer cyber threats.

References

- [1] S. Mali et al., "Federated Reinforcement Learning-Based Dynamic Resource Allocation and Task Scheduling in Edge for IoT Applications," *Sensors (Basel, Switzerland)*, vol. 25, no. 7, p. 2197, 2025.
- [2] S. N. Makhadmeh, S. Fraihat, M. Awad, Y. Sanjalawe, M. A. Al-Betar, and M. A. Awadallah, "A crossover-integrated Marine Predator Algorithm for feature selection in intrusion detection systems within IoT environments," *Internet of Things*, vol. 31, p. 101536, 2025.
- [3] M. J. Alenazi, M. A. Al-Khasawneh, S. Rahman, and Z. B. Faheem, "Deep Reinforcement Learning Based Flow Aware-QoS Provisioning in SD-IoT for Precision Agriculture," *Computational Intelligence*, vol. 41, no. 1, p. e70023, 2025.
- [4] A. Sharma, K. J. Singh, D. S. Kapoor, K. Thakur, and S. Mahajan, "Advancements in 5G-IoT Technology-Based Healthcare: A Focus on Healthcare Monitoring for Smart Cities," in *Mobile Crowdsensing and Remote Sensing in Smart Cities*: Springer, 2024, pp. 73-84.
- [5] T. M. Ghazal, "An IoT-Based Active Learning Convolutional Neural Networks Model for Predicting River Water Quality," *Journal of Soft Computing and Data Mining*, vol. 6, no. 1, pp. 319-332, 2025.
- [6] M. A. Almaiah, L. M. Saqr, L. A. Al-Rawwash, L. A. Altellawi, R. Al-Ali, and O. Almomani, "Classification of Cybersecurity Threats, Vulnerabilities and Countermeasures in Database Systems," *Computers, Materials & Continua*, vol. 81, no. 2, 2024.
- [7] M. Alkhawatrah, M. AlAyyad, and O. Korostynska, "Balanced inter-relay charging buffer-aided IOT networks," *EURASIP Journal on Wireless Communications and Networking*, vol. 2025, no. 1, p. 88, 2025.
- [8] A. Saaidah, O. Almomani, L. Al-Qaisi, and M. K. Madi, "An efficient design of RPL objective function for routing in internet of things using fuzzy logic," *Int. J. Adv. Comput. Sci. Appl.*, vol. 10, no. 8, pp. 184-190, 2019.
- [9] A. A. Almuqren, "Cybersecurity threats, countermeasures and mitigation techniques on the IoT: Future research directions," *Journal of Cyber Security and Risk Auditing*, vol. 1, no. 1, pp. 1-11, 2025.
- [10] A. Z. Mohammed Nafa, A. A. Obed, A. J. Abid, S. J. Yaqoob, M. Bajaj, and M. Shabaz, "Sensorless real-time solar irradiance prediction in grid-connected PV systems using PSO-MPPT and IoT-enabled monitoring," *Energy Informatics*, vol. 8, no. 1, p. 99, 2025.
- [11] A. ALDabbas, L. B. Baniata, B. A. AlSaaidah, Z. Mustafa, M. Alali, and R. Rateb, "Artificial intelligence-driven method for the discovery and prevention of distributed denial of service attacks," *Int J Artif Intell ISSN*, vol. 2252, no. 8938, p. 8938, 2025.
- [12] T. Alsalem and M. Amin, "Towards Trustworthy IoT Systems: Cybersecurity Threats, Frameworks, and Future Directions," *Journal of Cyber Security and Risk Auditing*, vol. 2023, no. 1, pp. 3-18, 2023.
- [13] S. R. Addula and A. Ali, "A Novel Permissioned Blockchain Approach for Scalable and Privacy-Preserving IoT Authentication," *Journal of Cyber Security and Risk Auditing*, vol. 2025, no. 4, pp. 222-237, 2025.
- [14] N. K. Al-Okbi et al., "The Intersection of AI and the Internet of Things (IoT): Transforming Data into Intelligence," *Mastering the Minds of Machines: A Journey into Deep Learning and AI*, p. 149, 2025.
- [15] R. Alabdallat, M. Abualhaj, and A. Abu-Shareha, "Android Malware Detection Using a Modified Dwarf Mongoose Algorithm," *International Journal of Intelligent Engineering & Systems*, vol. 18, no. 8, 2025.
- [16] R. D. Alabdallat, M. M. Abualhaj, and A. Abu-Shareha, "Enhanced Multiclass Android Malware Detection Using a Modified Dwarf Mongoose Algorithm," *International Journal of Analysis and Applications*, vol. 23, pp. 248-248, 2025.
- [17] L. Abualigah and A. Smerat, "Future Directions in Artificial Intelligence: Trends, Challenges, and Human Implications," in *Mastering the Minds of Machines*: CRC Press, 2025, pp. 194-200.
- [18] S. Jaradat, N. Acharya, S. Shivshankar, T. I. Alhadidi, and M. Elhenawy, "AI for Data Quality Auditing: Detecting Mislabelled Work Zone Crashes Using Large Language Models," *Algorithms*, vol. 18, no. 6, p. 317, 2025.
- [19] A. Arabiat and M. Altayeb, "Driving behavior analytics: an intelligent system based on machine learning and data mining techniques," *Bulletin of Electrical Engineering and Informatics*, vol. 14, no. 3, pp. 2055-2065, 2025.
- [20] S. Abuowaida et al., "Evidence Detection in Cloud Forensics: Classifying Cyber-Attacks in IaaS Environments using machine learning," 2025.
- [21] O. Almomani et al., "A Robust Model for Android Malware Detection via ML and DL classifiers," *Mesopotamian Journal of Big Data*, vol. 2025, pp. 261-277, 2025.
- [22] O. Almomani, A. Alsaaidah, A. A. Abu-Shareha, A. Alzaqebah, M. A. Almaiah, and Q. Shambour, "Enhance URL Defacement Attack Detection Using Particle Swarm Optimization and Machine Learning," *Journal of Computational and Cognitive Engineering*, vol. 4, no. 3, 2025.

- [23] W. Ibrar, D. Mahmood, A. S. Al-Shamayleh, G. Ahmed, S. Z. Alharthi, and A. Akhunzada, "Generative AI: a double-edged sword in the cyber threat landscape," *Artificial Intelligence Review*, vol. 58, no. 9, p. 285, 2025.
- [24] O. Alheyasat, "Web Phishing Detection and Awareness Utilizing Hybrid Machine Learning Algorithms," *International Journal of Advances in Soft Computing & Its Applications*, vol. 17, no. 2, 2025.
- [25] M. M. Abualhaj, S. N. Al-Khatib, M. Al Zyoud, I. Qaddara, and M. Anbar, "Enhancing Intrusion Detection System Performance Using a Hybrid of Harris Hawks and Whale Optimization Algorithms," *Engineering, Technology & Applied Science Research*, vol. 15, no. 4, pp. 24354-24361, 2025.
- [26] M. M. Abualhaj, S. Al-Khatib, A. Alalousi, M. O. Hiari, and M. Al Zyoud, "Phishing Attack Detection Using Whale Optimization Algorithm-Based Feature Selection," in *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)*, 2025: IEEE, pp. 1-6.
- [27] R. Masadeh et al., "Narwhal Optimizer: A Nature-Inspired Optimization Algorithm for Solving Complex Optimization Problems," *Computers, Materials & Continua*, vol. 85, no. 2, pp. 3709--3737, 2025. [Online]. Available: <http://www.techscience.com/cmc/v85n2/63798>.
- [28] M. A. Shenify, A. S. Alghamdi, and A. F. Alharthi, "Hybrid Supervised Machine Learning-based Intrusion Detection System of Internet of Things," *International Journal of Advances in Soft Computing & Its Applications*, vol. 16, no. 2, 2024.
- [29] A. Q. Algaolahi, A. A. Hasan, A. Sallam, A. M. Sharaf, A. A. Abdu, and A. A. Alqadi, "Port-scanning attack detection using supervised machine learning classifiers," in *2021 1st International Conference on Emerging Smart Technologies and Applications (eSmarTA)*, 2021: IEEE, pp. 1-5.
- [30] Q. A. Al-Haija, E. Saleh, and M. Alnabhan, "Detecting port scan attacks using logistic regression," in *2021 4th International symposium on advanced electrical and communication technologies (ISAECT)*, 2021: IEEE, pp. 1-5.
- [31] M. Aamir, S. S. H. Rizvi, M. A. Hashmani, M. Zubair, and J. Ahmad, "Machine learning classification of port scanning and DDoS attacks: A comparative analysis," *Mehran University Research Journal Of Engineering & Technology*, vol. 40, no. 1, pp. 215-229, 2021.
- [32] A. Kumari and I. Sharma, "Securing the internet of things using AI-enabled detection of attacks via port scans in IoT networks," in *2023 International Conference on Power Energy, Environment & Intelligent Control (PEEIC)*, 2023: IEEE, pp. 348-352.
- [33] J.-B. Altidor and C. Talhi, "Enhancing Port Scan and DDoS attack detection using genetic and machine learning algorithms," in *2024 7th Conference on Cloud and Internet of Things (CIoT)*, 2024: IEEE, pp. 1-7.
- [34] Z. Yan, B. Wang, Z. Chen, Y. Zeng, X. Wang, and Q. Wen, "Port Scanning Detection Method Based on Ensemble Learning," in *2024 7th International Conference on Mechanical, Control and Computer Engineering (ICMCCE)*, 2024: IEEE, pp. 163-167.
- [35] O. Z. Akif, S. M. Ali, A. F. Sabih, A. T. Sadiq, and S. Subramaniam, "Intrusion Detection System for IoT Based on Modified Random Forest Algorithm," *Iraqi Journal for Computer Science and Mathematics*, vol. 6, no. 2, p. 17, 2025.
- [36] A. Alsaaidah, O. Almomani, A. A. Abu-Shareha, M. M. Abualhaj, and A. Achuthan, "ARP spoofing attack detection model in IoT network using machine learning: Complexity vs. accuracy," *Journal of Applied Data Sciences*, vol. 5, no. 4, pp. 1850-1860, 2024.
- [37] H. Al-Mimi, N. A. Hamad, M. M. Abualhaj, M. S. Daoud, A. Al-Dahoud, and M. Rasmi, "An Enhanced Intrusion Detection System for Protecting HTTP Services from Attacks," *International Journal of Advances in Soft Computing & Its Applications*, vol. 15, no. 3, 2023.
- [38] Y. Sanjalawe, S. Fraihat, S. Al-E'mari, M. Abualhaj, S. Makhadmeh, and E. Alzubi, "Smart load balancing in cloud computing: Integrating feature selection with advanced deep learning models," *Plos one*, vol. 20, no. 9, p. e0329765, 2025.
- [39] M. M. Abualhaj, S. N. Al-Khatib, M. Al-Zyoud, I. Qaddara, M. O. Hiari, and S. M. A. Aldossary, "Enhanced Network Communication Security Through Hybrid Dragonfly-Bat Feature Selection for Intrusion Detection," *Journal of Communications*, vol. 20, no. 5, 2025.
- [40] M. Abualhaj, Q. Shambour, A. Abu-Shareha, S. Al-Khatib, and A. Amer, "Enhancing malware detection through self-union feature selection using gray wolf optimizer," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 37, no. 1, p. 197, 2025.
- [41] M. Madi, F. Jarghon, Y. Fazea, O. Almomani, and A. Saaidah, "Comparative analysis of classification techniques for network fault management," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 28, no. 3, pp. 1442-1457, 2020.
- [42] H. A. Al-Ofeishat, J. S. Alkasassbeh, A. Awajan, and M. Alazab, "Analysis and Comparison of Raw Network Packet Datasets Using Machine Learning Classification and Grey Wolf Optimization," *International Journal of Advances in Soft Computing & Its Applications*, vol. 17, no. 1, 2025.
- [43] Y. A. Maz, M. Anbar, S. Manickam, M. M. Abualhaj, S. A. Almalki, and B. A. Alabsi, "Transfer Learning-Based Approach with an Ensemble Classifier for Detecting Keylogging Attack on the Internet of Things," *Computers, Materials and Continua*, vol. 85, no. 3, pp. 5287-5307, 2025.
- [44] Q. Shambour, M. Al-Zyoud, and O. Almomani, "Quantum-Inspired Hybrid Metaheuristic Feature Selection with SHAP for Optimized and Explainable Spam Detection," *Symmetry*, vol. 17, no. 10, p. 1716, 2025.